

$$f(x) = |\sin(x)|, \quad -\pi < x < \pi$$

$$\pi a_0 = \int_{-\pi}^{\pi} f(x) dx = 2 \int_0^{\pi} |\sin(x)| dx = 2 \cdot (-\cos(x)) \Big|_0^{\pi} = 2 \cdot (1 - (-1)) = 4$$

$$4 = 2 \cdot (+1) \cdot ((+1) + (-1)) = 4 \Rightarrow \boxed{a_0 = \frac{4}{\pi}}$$

$$\pi a_n = \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \int_{-\pi}^{\pi} \underbrace{|\sin(x)|}_{\text{PAR}} \underbrace{\cos(nx)}_{\text{PAR}} dx = 2 \int_0^{\pi} \underbrace{\sin(x)}_P \underbrace{\cos(nx)}_{V'} dx$$

Integração por partes

$$\Rightarrow \frac{\pi}{2} a_n = \frac{\sin(x) \sin(nx)}{n} \Big|_0^{\pi} - \int_0^{\pi} \underbrace{\cos(x)}_P \underbrace{\sin(nx)}_{V'} dx$$

$$= -\frac{\cos(x)(-\cos(nx))}{n^2} \Big|_0^{\pi} + \frac{1}{n^2} \int_0^{\pi} \sin(x) \cos(nx) dx$$

$$= -\frac{[+1(+\cos(n\pi)) + 1(+1)]}{n^2} + \frac{1}{n^2} \int_0^{\pi} \sin(x) \cos(nx) dx = -\frac{(1+(-1)^n)}{n^2} + \frac{1}{n^2} \frac{\pi a_n}{2}$$

$$\Rightarrow \frac{\pi a_n}{2n^2} - \frac{\pi a_n}{2n^2} = a_n \left(\frac{\pi}{2n^2} \right) (n^2 - 1) = \frac{-1 - (-1)^n}{n^2} \Rightarrow \int_0^{\pi} \sin(x) \cos(nx) dx = \frac{\pi a_n}{2}$$

$$\Rightarrow \boxed{a_n = -\frac{2}{\pi} \frac{(-1)^n + 1}{n^2 - 1}}$$

$$\pi b_n = \int_{-\pi}^{\pi} \underbrace{|\sin(x)|}_{\text{PAR}} \underbrace{\sin(nx)}_{\text{IMPAR}} dx = 0$$

$$\begin{aligned} P &= \sin(x) \\ V &= \sin(nx)/n \\ (PV)' &= P'V + PV' \\ PV \Big|_a^b &= \int_a^b P'V + \int_a^b PV' \\ PV' &= PV - P'V \\ \int_0^{\pi} \sin(x) \cos(nx) dx &= \frac{\sin(x) \sin(nx)}{n} \Big|_0^{\pi} - \int_0^{\pi} \cos(x) \sin(nx) dx \end{aligned}$$

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1) b)